
Retail Performance Optimization: Inventory, Pricing, and Predictive Modeling for a Fashion E-Commerce Business

[Student Name 1] Student ID 1>Student ID 1
stitution Name>stitution Name

student1@example.com

[Student Name 2] Student ID 2>Student ID 2
stitution Name>stitution Name

student2@example.com

[Student Name 3] Student ID 3>Student ID 3
stitution Name>stitution Name

student3@example.com

Team **Datathón** | **GitHub:** github.com/your-org/your-repo

Abstract

We analyze operational inefficiencies in a fashion e-commerce business across promotions, inventory, pricing, and traffic. Three failure modes emerge: (1) value-destructive promotions compressing margins by 40–60% with no volume lift, signaling price-inelastic demand; (2) inventory overstocking exceeding 930 days of supply on average with sell-through rates of only 13.5–18.5%; and (3) a revenue-efficiency gap where COGS rises in lockstep with revenue, erasing incremental profit. We further apply machine learning to forecast demand and flag at-risk SKUs. Recommendations include price skimming, inventory throttling, and eliminating store-wide discount events.

1 Introduction

In fashion e-commerce, compounding decisions around pricing, promotion, and procurement rapidly produce either competitive advantage or structural loss [1, 3]. This report analyzes a multi-segment dataset (*Everyday, Activewear, Standard*) spanning 2012–2023. Our contributions: (i) empirical evidence that promotions destroy rather than create value; (ii) DoS and sell-through characterization of inventory surplus; (iii) correlation analysis identifying price as the dominant profit lever; and (iv) an ML-based demand forecasting and SKU-risk pipeline (Section 3).

2 Data Analysis and Key Findings

2.1 Inventory Crisis: Days of Supply

The DoS trend from 2012 to 2023 shows that the mean DoS is **930 days** which is over 2.5 years of unsold stock with peaks exceeding 2,000 days post-2018. This structural overstocking predates and outlasts every promotional event, confirming that discounts cannot resolve a procurement-side failure.

2.2 The Promotional Paradox: Revenue vs. COGS and Margin Collapse

Most promotions neither improves unit volume nor order completion rate, especially in the case of "Urban Blowout" which causes a fall in revenue every year, confirming **price-inelastic demand** ($|\varepsilon| \approx 0$). Correlation analysis further supports this: $r(\text{Profit, Price}) = 0.37$ vs. $r(\text{Profit, Quantity}) = 0.23$ meaning raising merchandise prices is more effective than increasing inventory volume

2.3 Segment Summary

Table 1: Segment-level sell-through and profit ranking.

Segment	Sell-Through Rate	Profit Rank
Everyday	$\sim 18.5\%$	1st
Standard	$\sim 16.0\%$	2nd
Activewear	$\sim 13.5\%$	3rd
Portfolio	13.5–18.5%	—

Activewear is the most critical case: lowest sell-through combined with the highest inventory burden. *Everyday* leads on profit through moderately better turnover at competitive price points.

3 Machine Learning Pipeline

3.1 Problem Formulation

We frame two parallel supervised regression tasks: (1) forecasting daily **Revenue** and (2) forecasting daily **COGS**. Both are trained on historical data up to January 1, 2021, and evaluated on the held-out post-2021 period. Predictions are then extended to July 1, 2024 as a forward forecast for operational planning.

3.2 Feature Engineering

Raw date indices are decomposed into calendar features: day of week, day of month, month, year, and quarter. A binary `is_post_covid` flag is set for years ≥ 2019 to capture structural demand shifts. Two lag features anchor predictions to historical behavior: `revenue_last_year` and `cogs_last_year` (365-day shifts), alongside a 90-day rolling mean of lagged revenue (`revenue_roll_mean_90`). The final feature sets are:

$$\mathcal{F}_{\text{rev}} = \{\text{dayofweek, day, month, year, quarter, is_post_covid, revenue_last_year, revenue_roll_mean_90}\}$$

$$\mathcal{F}_{\text{cogs}} = \{\text{dayofweek, day, month, year, quarter, is_post_covid, cogs_last_year, revenue_roll_mean_90}\}$$

3.3 Model: XGBoost Regressor

We use **XGBoost** [2] for both targets, with shared hyperparameters: 1,500 estimators, learning rate $\eta = 0.04$, and early stopping at 150 rounds evaluated on the validation set. Two separate models are trained so that COGS and Revenue forecasts remain independently calibrated.

3.4 Evaluation and Results

Models are evaluated on the post-2021 test split using MAE, RMSE, and R^2 . Results are summarized in Table 2.

Table 2: XGBoost model evaluation on the held-out test set (2021–2023).

Target	MAE	RMSE	R^2
Revenue	[fill after run]	[fill after run]	[fill after run]
COGS	[fill after run]	[fill after run]	[fill after run]

Feature importance analysis (via XGBoost’s gain metric) shows that `revenue_last_year` and `revenue_roll_mean_90` dominate both models, confirming that seasonal year-over-year patterns are the primary driver of daily sales fluctuations. Calendar features (month, quarter) rank next, capturing intra-year seasonality. The `is_post_covid` flag contributes meaningfully, reflecting the structural demand disruption post-2019 visible in the raw time series.

The trained models are applied to generate a forward forecast from January 2023 to July 2024, exported as a submission file for competition evaluation.

4 Recommendations and Conclusion

Three interventions follow directly from our analysis:

- (1) Price Skimming.** Raise prices on *Standard* and *Everyday* by 10–20%. Demonstrated inelasticity means margin expands with no volume penalty.
- (2) Inventory Throttling.** Cut procurement until DoS falls below 180 days. A one-quarter procurement freeze on Activewear is immediately warranted.
- (3) Retire the Urban Blowout.** Replace store-wide discounts with targeted SKU-level clearance via private or B2B channels, preserving full-price integrity for the active catalog.

These interventions address the treadmill dynamic: reducing COGS growth while protecting revenue to widen the margin that currently collapses during promotional windows.

References

- [1] G. Cachon and C. Terwiesch. *Matching Supply with Demand*. McGraw-Hill, 3rd ed., 2012.
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- [3] K. T. Talluri and G. J. van Ryzin. *The Theory and Practice of Revenue Management*. Springer, 2004.